Programming Fundamentals
Lab-05

Problem 01:

Write a function that takes a number as input.

- If the number has exactly **6 digits**, the function should return the **sum of all its digits**.
- If the number is **not 6 digits**, the function should return **-1**.

Also, write int main to test it.

```
int DigitsSum(int Number)
{
        // Write the function here
}
int main()
{
        int Num;
        cout<<"Enter the Six Digit Number: ";
        cin>>Num;
        // Write the code here...!!!
}
```

Sample Input: 153426
Sample output: 21

Problem 02:

Write a program that takes a **6-digit number** from the user.
The program should print each digit in **words** (like ONE, TWO, THREE...) on the screen.Also, write int main to test it.

Sample Input: 253426
Sample output: TWO FIVE THREE FOUR TWO SIX

Problem 03:

Write a function that takes a number as input and checks if it is **even** or **odd**.

- If the number is even, return true.
- If the number is odd, return false.

A number is even if `number % 2 == 0`. Otherwise, it's odd.

Then, write a `main` function to test this.

Sample Input: 426
Sample output: Even Number

Sample Input: 111
Sample output: Odd Number

Problem 04:

Write a function that takes 5 input integers as parameters and returns the maximum from these integers. Also, write int main to test it.

Sample Input: 60 41 82 93 51
Sample Output: 93

Problem 05:

Programming Fundamentals (PF) has **6 sections**.
Each section has an **average score**.

You need to:

1. Write a **function** that takes the 6 averages as input.
2. The function should find out **which section has the highest average** and return that section number.
3. Write a `main` function to test this program.

Sample Input:	**Sample Output:**
B 90 D 80 C 60 A 99 E 91 F 80	A got the highest average *NOTE: The order of the sections could be random.* ***(Bonus)*** *how can you write a program using a function (or functions) which should display the following output* **A** got the highest average **of 99**

Problem 06:

Write a function which takes a character as input and tells whether that character is Capital letter or Small letter or any other letter(1,2,3) Better write three functions **bool IsCapital()**, **bool IsSmall()**, and **int LetterType()**. Also, Write int main to test this function and fulfill the following input and output requirements.

Sample Input: A
Output: Capital letter
Sample Input: !
Sample Output: None

Sample Input: g
Output: Small letter

Problem 07:

Write a Program that takes 4 points as input and tells whether these points are the coordinates of a Rectangle, Square, or Quadrilateral.

NOTE: Make separate functions to check whether it is rectangle/square/quadrilateral. i-e (isRectangle, isSquare, isQuadrilateral)

Sample Input:

P1 0 0
P2 1 0
P3 1 1
P4 0 1

Output:

It's a square

Also, Write int main to test this program.

Problem 08:

Make the following two functions:

- Write a function Digit2Word which takes as a parameter one digit and outputs a digit in words like if 1 is passed it should print "One", if 2 prints "Two" and so on, on the console screen.
- Write a Function that will take at max: a six-digit number and output each of its digits in words,
- **(Challenge) if the number is less than 6 digit it should not print initial zeros. If the number is greater than 6 digits then it should output the wrong input. Also write int main to test it.**

Sample Input: 651432
Output: Six Five One Four Three two
(Challenge)

Sample Input: 1432
Sample Output: One Four Three Two.

Problem 09:

ITU hired you to implement their grading policy given below. Your responsibility is to write the function which takes marks as a parameter and returns the respective letter Grade. Also write **int main** to test it.

If a student got marks

>= 90 or <= 100	**Grade = A**
>= 80 or < 90	**Grade = B**
>= 70 or < 80	**Grade = C**
>= 60 or < 70	**Grade = D**
>= 50 or < 60	**Grade = E**
Less than 50	**Grade = F**

Sample Input: Marks = 83
Sample output: Grade = B

Sample Input: Marks = 70.8
Sample output: Grade = C

Problem 10:

Write a Function **Calculate()** that simulates a simple calculator. It takes two integers and a character as a parameter. If the character is a '+', the sum is returned; if it is a '-', the difference is returned; if it is a '*', the product is returned; if it is a '/', the quotient is returned; and if it is a '%', the remainder is returned.

Then write a **calculatorDisplay()** function which takes two integers and a character and outputs the corresponding result (by calling **Calculate()** function). Then write **int main()** to test this program.

Sample Input: 12%7
Sample Output: 5

Sample Input: 19x10
Sample Output: 190

```
O===O
 \O/
  |
 /\
```

Happy Coding... :)
Good luck